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Systems and methods for redeeming rewards for cash at an ATM

Abstract

An automated teller machine (ATM) includes a network interface, an input/output device configured to exchange data with a customer, and a processing circuit comprising a processor and memory. The memory is structured to store instructions that are executable by the processor and cause the processing circuit to receive account information identifying a credit account associated with the customer, transmit the account information identifying the credit account, receive reward information associated with the credit account, display a reward indication notifying the customer of reward information and providing the customer with an ability to indicate a preference to perform a currency reward withdrawal, receive data comprising an indication of the preference of the customer to perform the currency reward withdrawal, transmit a redemption request based on the indicated preference, receive a redemption request authorization, and dispense an amount of currency corresponding to an amount identified in the redemption request authorization.

Inventors: Collins; Megan (Westfield, NJ), Phelps; Sarah W. (Annapolis, MD), Rogers; Matthew S. (Hoboken, NJ)

Applicant: Wells Fargo Bank, N.A. (San Francisco, CA)

Family ID: 69057540

Assignee: Wells Fargo Bank, N.A. (San Francisco, CA)

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Primary Examiner: Refai; Sam

Attorney, Agent or Firm: Foley & Lardner LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/398,705 filed Aug. 10, 2021, which is a continuation of U.S. patent application Ser. No. 16/562,041 filed Sep. 5, 2019, which is a divisional of U.S. patent application Ser. No. 15/393,882 filed Dec. 29, 2016, now U.S. Pat. No. 10,521,814 issued Dec. 31, 2019, each of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) The present invention relates generally to the field of delivery of financial rewards to customers of a financial institution.

BACKGROUND

(2) Many issuers of credit cards offer rewards to customers. For example, some credit card issuers provide rewards for every dollar spent on a credit card. Accordingly, as a customer continues to use a rewards credit card, the customer accumulates rewards. After the customer has earned rewards, the customer can then redeem the rewards. Some financial institutions restrict how the accumulated rewards can be redeemed. For example, some rewards can only be redeemed in certain types of transactions (e.g., for certain types of products). In this regard, financial institutions and customers of those institutions may benefit from a system that enables customers to redeem financial rewards in a more convenient manner.

SUMMARY

(3) A first example embodiment relates to an automated teller machine (ATM). The ATM includes a network interface, an input/output device configured to exchange data with a customer, and a processing circuit comprising a processor and memory. The memory is structured to store instructions that are executable by the processor and cause the processing circuit to receive, through the input/output device, credit account information identifying a credit account associated with the customer, transmit, through the network interface, the credit account information identifying the credit account, receive, through the network interface, financial reward information associated with the credit account, display a reward indication, through the input/output device,

notifying the customer of customer reward information and providing the customer with an ability to indicate a preference to perform a currency reward withdrawal, receive data, through the input/output device, comprising an indication of the preference of the customer to perform the currency reward withdrawal, transmit, through the network interface, a financial reward redemption request based on the indicated preference, receive, through the network interface, a reward redemption request authorization, and dispense, through the input/output device, an amount of currency corresponding to an amount identified in the reward redemption request authorization.

(4) In various arrangements, the indicated customer preference includes a portion of a financial reward amount identified in the financial reward information. In some arrangements, the ATM is further configured to receive data indicative of an interaction of the customer with the ATM through the input/output device and initiate a customer ATM session responsive to the data indicative of the interaction. The data indicative of the interaction can be tokenized and received from a computing device associated with the customer. The data indicative of the interaction can include an account number associated with the credit account.

(5) In some arrangements, the ATM is configured to display an instruction for an input comprising authentication information, receive the authentication information, and generate a transaction request comprising the authentication information and the credit account information.

(6) In some arrangements, the ATM is further configured to present a plurality of buttons. The plurality of buttons can include a reward redemption button structured to commence a reward redemption process, a reward balance inquiry structured to present financial award information, and a register additional cards button structured to commence a credit account registration process. In some arrangements, the ATM is further configured to register the credit account associated with the customer.

(7) Another example embodiment relates to a computer-implemented method. The method includes receiving, by an automated teller machine (ATM), credit account information identifying a credit account associated with a customer, transmitting, by the ATM, the credit account information to a financial institution computing system, receiving, by the ATM, financial reward information associated with the credit account, displaying, by the ATM, a reward indication notifying the customer of the financial reward information and providing the customer with the ability to indicate a preference to perform a currency reward withdrawal, receiving, by the ATM, data comprising an indication of the preference of the customer to perform the currency reward withdrawal, transmitting, by the ATM, a financial reward redemption request based on the indicated preference, receiving, by the ATM, a reward redemption request authorization, and dispensing, by the ATM, an amount of currency corresponding to an amount identified in the reward redemption request authorization. In some arrangements, receiving, by the ATM, information involves receiving an account number associated with the credit account.

(8) In some arrangements, the method includes displaying, by the ATM, an instruction for an input comprising authentication information, receiving, by the ATM, the authentication information, and generating, by the ATM, a transaction request comprising the authentication information and the credit account information.

(9) In some arrangements, receiving, by the ATM, data comprising an indication of preference of customer involves receiving an indication of an amount of earned awards for redemption.

(10) In various arrangements, the method further comprises generating a reward redemption request associated with the amount of earned awards for redemption.

(11) In some embodiments, the method further includes receiving, by the ATM, data indicative of an interaction of the customer with the ATM through an input/output device and initiating, by the ATM, a customer ATM session responsive to the data indicative of the interaction. Receiving the data indicative of the interaction can include receiving, by the ATM, the data from a computing device associated with the customer and tokenizing, by the ATM, the data.

(12) Yet another implementation of the present disclosure is a computer-implemented method. The

method includes receiving, by a financial institution (FI) computing system, registration input comprising credit account information identifying a credit account and customer credentials associated with a customer, authenticating, by the FI computing system, the customer based on the customer credentials, transmitting, by the FI computing system, a registration interface responsive to the customer authentication, receiving, by the FI computing system, customer registration information, and associating, by the FI computing system, the received customer registration information with the credit account information.

(13) In some arrangements, the method further involves determining, by the FI computing system, that the credit account was not previously registered based on the credit account information.

(14) In some arrangements, the method further involves determining, by the FI computing system, that the credit account was previously registered based on the credit account information and transmitting, by the FI computing system, a notification.

(15) In some arrangements, the method further includes identifying, by the FI computing system, one or more additional credit accounts associated with the customer and transmitting, by the FI computing system and for each of the one or more additional credit accounts, the registration interface.

(16) In various arrangements, the method further includes receiving, by the FI computing system, data indicative of an interaction of the customer with an automated teller machine and initiating, by the FI computing system, a customer session responsive to the data indicative of the interaction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram of an environment of a customer according to an example embodiment.

(2) FIG. 2 is a flow diagram of a method for exchanging customer credit rewards for cash at an ATM according to an example embodiment.

(3) FIG. 3 is a flow diagram of a method for registering a customer credit card for an ATM cash redemption program according to an example embodiment.

(4) FIGS. 4A-4D are example screens generated on a graphical interface at an ATM, according to example embodiments.

DETAILED DESCRIPTION

(5) Referring generally to the figures, various systems, methods, and apparatuses related to a financial reward redemption system structured to enable credit-only customers of a financial institution to withdraw financial rewards for cash via an Automated Teller Machine (ATM) are described.

(6) According to various example embodiments, as described in further detail below, facilitating the cash withdrawal of financial rewards by credit-only customers via an ATM may incentivize credit-only customers to use a payment account associated with a financial institution to earn financial rewards. Unlike traditional arrangements, it is not necessary for the customer **110** to have a checking, debit, or any other type of account to have access to the ATM. Instead, using the system described herein, the credit-only customer can conveniently make cash withdrawals on a reward account associated with a credit card at an ATM. Beneficially, this system provides customer with convenient access to earned financial rewards.

(7) In addition, embodiments described herein solve the technical problem of distributing cash rewards earned through a customer credit account via an ATM. This is addressed by associating customer ATM authentication data to customer credit financial reward information stored in a database associated with a financial institution computing system. This way, the customer can have access to ATMs associated with the financial institution to redeem a financial reward when the

customer does not have any other account with the financial institution.

(8) An example implementation may be described as follows. A customer approaches an ATM and inserts a credit card associated with a credit account held by the customer at a financial institution into the ATM. The contemplated financial reward redemption system is configured to determine whether the credit account has been registered for an ATM cash redemption program. Responsive to determining that the credit account is not registered, the contemplated financial reward redemption system is configured to enable the customer to register for the ATM cash redemption program. During the registration process, the customer is presented with a registration interface on an ATM that prompts the customer to input authentication information (e.g., a birthdate, social security number, or the like). The contemplated financial reward redemption system is configured to authenticate the customer based on the input information. The contemplated reward redemption system is further configured to enable the customer to establish credentials (e.g., a PIN) for the credit account. Thus, later on, the customer may approach an ATM, insert the credit card, and input the PIN. The contemplated financial reward redemption system is configured to authenticate the customer using the PIN, retrieve financial reward information associated with the credit account, and present the customer with a display giving the customer the ability to indicate a preference to redeem a financial reward for currency. Responsive to receiving such a preference, the contemplated financial reward redemption system is configured to dispense an amount currency that corresponds to a selection provided by the customer at the ATM.

(9) Referring to FIG. 1, a block diagram of a reward redemption system **100** is shown according to an example embodiment. As will be described in further detail below, the reward redemption system **100** facilitates a customer **110** redeeming financial rewards associated with a credit card **112** for cash at an ATM **122**. A financial institution computing system **130** receives information from the customer **110** via the ATM **122** to authenticate the customer **110** and identify financial rewards earned by the customer **110**. The financial institution computing system **130** also converts an amount of earned rewards to a monetary value and causes the ATM **122** to dispense an amount of currency corresponding to the monetary value.

(10) The customer **110** includes any customer of a financial institution associated with the financial institution computing system **130** who is or may be eligible to participate in the financial rewards program offered by the financial institution. The customer **110** has at least one financial rewards account at the financial institution that may be associated with a credit card **112** held by the customer **110** at the financial institution.

(11) The reward redemption system **100** includes a card network computing system **140** associated with a card network, a financial institution computing system **130** associated with a financial institution, an ATM **122**, and a mobile device **114** associated with the customer, whereby these components are communicably coupled to each other over a network **150**. The network **150** provides communicable and operative coupling between the components described herein to provide and facilitate the exchange of communications (e.g., data, instructions, messages, values, commands, etc.). The network **150** is a data exchange medium, which may include wireless networks (e.g., cellular networks, Bluetooth®, WiFi, Zigbee®, etc.), wired networks (e.g., Ethernet, DSL, cable, fiber-based, etc.), or a combination thereof. In some arrangements, the network includes the Internet. In some arrangements, the network **150** may further include a proprietary banking network (e.g., between the financial institution computing system **130** and the ATM **122**) to provide secure or substantially secure communications.

(12) The card network computing system **140** is a computing system associated with a card network. Examples of card networks include Visa®, MasterCard®, etc. The card network computing system **140** performs operations associated with the generation and issuance of payment card tokens. Payment card tokens are surrogate values that replace the primary account number (“PAN”) associated with a payment card (e.g., the credit card **112**), such as a credit card, debit card, ATM card, stored value card, etc. Payment card tokens can pass basic validation rules of an account

number. Hence, in the case of a credit card, the payment card token for a given credit card “looks like” a real credit card number (e.g., a sixteen-digit number), but in fact is only a token. As part of a token generation process, steps are taken such that the generated payment card token does not have the same value as or otherwise conflicts with a real PAN (e.g., a real debit card number). A given payment card token may be provisioned to various locations for use in various types of scenarios, including ATMs for performing various financial operations, storage at a mobile device (e.g., a smartphone) for in-person or on-line transactions with a merchant, and so on.

(13) The card network computing system **140** includes a network interface circuit **142**, a token provisioning circuit **144**, and a token database **146**. The network interface circuit **142** enables the card network computing system **140** to exchange data over the network **150**. As such, the network interface circuit **142** allows the card network computing system **140** to exchange data to remote computing devices (e.g., the financial institution computing system **130**, the mobile device **114**, etc.). The token database **146** is a storage medium retrievably storing token information (e.g., a token vault), and may include non-transient data storage mediums (e.g., local disc or flash-based hard drives, local network servers, and the like) or remote data storage facilities (e.g., cloud servers).

(14) The token provisioning circuit **144** is configured to provision and manage tokens. In one aspect, the token provisioning circuit **144** can generate a new unique code to be provisioned as a token, and associate the token with a PAN (e.g., a PAN associated with the credit card **112** of the customer **110**). Hence, in some embodiments, payment card tokens are generated by the card network computing system **140** and payment token-to-PAN mapping information is maintained in the token database **146**. In another aspect, the token provisioning circuit **144** may be able to replace tokens as well as activate and deactivate tokens. The token provisioning circuit **144** may also be configured to associate permissions with each token, thereby allowing or disallowing the transmission or use of data associated with a given token.

(15) For example, when a financial institution issues a credit card **112** having a PAN to a customer **110** having a mobile device **114** with a mobile wallet client application **118** implemented thereon, the card network computing system **140** may assign a payment token corresponding to the PAN and the mobile device **114** associated with the customer **110**. In other words, the payment token can be used to identify the PAN and the mobile device **114**. The PAN, the payment token, and information relating to their respective associations may be stored in the token database **146**, thus enabling the customer **110** to engage in mobile wallet transactions using the mobile device, as will be described below.

(16) The financial institution computing system **130** is a computing system associated with a financial institution (e.g., a bank, a credit card issuer, etc.). The financial institution computing system **130** includes a customer database, a rewards management circuit **134**, a data exchange circuit **136**, and a network interface circuit **132**. The network interface circuit **132** is configured to allow the financial institution computing system **130** to exchange data over the network **150** (e.g., with the ATM **122**, the mobile device, **114**, and the card network computing system **140**).

(17) The customer database **138** is configured to retrievably store customer information relating to the various operations discussed herein, and may include non-transient data storage mediums (e.g., local disc or flash-based hard drives, local network servers, and the like) or remote data storage facilities (e.g., cloud servers). The customer database **138** includes personal customer information (e.g., names, addresses, phone numbers, and so on), identification information (e.g., PINs, social security numbers, driver's license numbers, biometric data, and so on), and customer financial information (e.g., account numbers, account balances, customer financial reward information, token information, available credit, credit history, transaction histories, and so on).

(18) The payment token-to-PAN mapping information generated and stored at the card network computing system **140** discussed above may also be maintained at the customer database **138** at the financial institution computing system **130**. As such, for example, the financial institution

computing system **130** may provide additional token-related management services to customers that are not available through the card network computing system **140**. Such services may be useful in situations where customers have multiple different types of accounts (e.g., multiple different types of credit cards, such that a single card network computer system does not have a global view of all of the payment tokens in existence for a given customer). Such services may be useful in situations where other information in addition to account numbers is tokenized by the financial institution computing system **130** or other computing systems, thereby creating tokens that are not payment tokens.

(19) The data exchange circuit **136** is configured to exchange data among the customer database **138**, the ATM **122**, the mobile device **114**, and the card network computing system **140**. In one aspect, the data exchange circuit **136** may be configured to exchange tokens and permissions with the card network computing system **140**, the ATM **122** (e.g., for authentication purposes) and the mobile device **114**. The data exchange circuit **136** may also be configured to facilitate transactions involving the ATM **122** and the mobile device **114**. In some arrangements, the data exchange circuit **136** is communicatively coupled to the mobile device **114** by an application (e.g., the mobile wallet client application **118** as will be described in greater detail below). Further, in some arrangements, the data exchange circuit **136** provides a payment token from the card network computing system **140** to the mobile wallet client application **118** over the network **150**. The data exchange circuit **136** is also configured to cooperate with other components of the financial institution computing system **130** to authenticate and authorize various requests received from the ATM **122**.

(20) The rewards management circuit **134** is configured to process reward payment requests from the customer **110**. In some arrangements, the rewards management circuit **134** is configured to perform all of the operations discussed above in relation to the data exchange circuit **136** in addition to performing other operations. Other operations that the rewards management circuit **134** is further configured to perform may include various account management functions, such as maintaining customer financial reward information in the customer database **138** (including, for example, updating reward balances and processing reward redemption requests by charging those requests to a rewards account). For example, when currency is redeemed by the customer **112** using the system described herein, the rewards management circuit **134** updates the associated rewards account by deducting from the account an amount of rewards that corresponds to the amount of currency redeemed by the customer **112** (e.g., the customer **112** redeems \$100 and the rewards management circuit **134** updates the rewards account to reflect \$100 worth of rewards less).

(21) The ATM **122** is a computing system configured to provide an interface between a customer **110** and the financial institution computing system **130** allowing the customer **110** to access information at and perform transactions with the corresponding financial institution. For example, the ATM **122** may be configured to allow the customer **110** to view financial account balances, deposit checks, transfer funds, or withdraw funds from a given account in the form of cash. In one embodiment, the ATM **122** is configured to allow a customer to view financial rewards that they have earned through a credit account at the financial institution. The ATM **122** may be disposed at a brick and mortar banking facility associated with the financial institution, or may be a standalone computing terminal (e.g., disposed at an unrelated retail facility, within an office building, etc.).

(22) The ATM **122** includes a network interface circuit **124** enabling the ATM **122** to exchange data over the network **150**, an ATM circuit **126**, and an ATM I/O device **128**. Similar to the mobile I/O device **120**, to be described below, the ATM I/O device **126** includes hardware and associated logics configured to enable the ATM **122** to exchange information with the customer **110** and the mobile device **114** (e.g., via corresponding hardware and logics at the mobile I/O device **120**). In some arrangements, the ATM I/O device **128** also includes hardware and associated logics that enable the ATM **122** to dispense paper currency (e.g., cash) to the customer **110**.

(23) In some arrangements, the ATM I/O device **128** includes a card reader **129**. The card reader **129** is configured to read a transaction card (e.g., the credit card **112**) provided to the ATM **122** by

the customer **110**. In some arrangements, the card reader **129** reads the transaction card through a customer swiping action. For example, the customer **110** swipes the transaction card by passing the magnetic strip of the card through the card reader **129**. In some arrangements, the card reader **129** reads the transaction card through a customer dipping action. For example, the customer **110** dips the transaction card by quickly inserting and removing the transaction card from the card reader **129**. In some arrangements, the card reader **129** is configured to read the transaction card through a customer inserting action. For example, the customer inserts the transaction card fully into the card reader **129** and the card remains inserted into the card reader **129** for the duration of a transaction sequence at the ATM **122**. In instances where the transaction card includes a smart chip, the customer **110** may be prompted to insert the card such that the card reader **129** can make contact with the smart chip for a period of time to read the smart chip. In some arrangements, the card reader **129** includes an NFC transceiver configured to receive transaction card information (e.g., a payment token) from the mobile device **114**.

(24) In some arrangements, the customer **110** inserts the credit card **112** into the card reader **129** of the ATM I/O device **128**, which reads the credit card **112** to obtain credit card information (e.g., an account number). The credit card information may then be communicated by the card reader **129** to the ATM circuit **126**, which, as will be described below, transmits the credit card information to the financial institution computing system **130**. The ATM circuit **126** is further configured to receive an authentication prompt from financial institution computing system **130** to be displayed (e.g., via a display included in the ATM I/O device **128**) on the ATM **122**. For example, the financial institution computing system **130** transmits an authentication prompt including a request for a PIN from the customer **110**. If the customer **110** (e.g., via a keyboard included on the ATM I/O device **128**) inputs the requested pin, the credit card **112** card is authenticated. Once the credit card **112** is authenticated, the customer **110** is authorized to complete reward transactions (e.g., a currency reward withdrawal) at the ATM **122** using the credit card **112**, as described further herein.

(25) The ATM circuit **126** is configured to enable the customer **110** to interface with the financial institution computing system **130**. The ATM circuit **126** exchanges information between the customer **110** via the mobile device **114**, the financial institution computing system **130**, and the card network computing system **140**. For example, the ATM circuit **126** may be configured to collect authentication information (e.g., a token and a PIN) from the mobile device **114** through a mobile wallet client application **118** as will be described below, and send the authentication information to the financial institution computing system **130** and/or the card network computing system **140** over the network **150**. Further, the ATM circuit **126** may be configured to receive an authentication confirmation from the financial institution computing system **130** and allow the customer to create financial reward operation requests. Such financial reward operation requests may include, for example, reward balance inquiries, cash reward withdrawal requests, and so on. The ATM circuit **126** assembles and transmits a customer's desired financial reward operation request to the financial institution computing system **130** which executes the corresponding financial operation. In some arrangements, the ATM circuit **126** ultimately provides the customer with a requested amount of cash in exchange for any financial rewards that the customer has earned through utilization of a credit account with the financial institution. In some arrangements, the ATM circuit **126** ultimately provides the customer with requested financial information (e.g., an account balance). As one of skill in the art would recognize, the ATM circuit **126** may provide the customer with various types of outputs corresponding to information exchanged with the financial institution computing system **130**.

(26) The mobile device **114** is a computing system associated with the customer **110**. The mobile device **114** includes any type of portable computing device that the customer **110** may use to communicate information with the financial institution computing system **130**. In this regard, the mobile device **114** may include any wearable or non-wearable device. Wearable devices refer to any type of device that an individual wears including, but not limited to, a watch (e.g., smart

watch), glasses (e.g., eye glasses, sunglasses, smart glasses, etc.), bracelet (e.g., a smart bracelet), etc. The mobile device **114** may also include a phone (e.g., smart phone, etc.), tablet, personal digital assistant, and/or laptop computer.

(27) In the example embodiment shown, the mobile device **114** includes a network interface circuit **116** enabling the mobile device **114** to exchange information over the network, a mobile wallet client application **118**, and a mobile input/output (“I/O”) device **120**. The mobile I/O device **120** includes hardware and associated logics configured to enable the mobile device **114** to exchange information with the customer **110**, the financial institution computing system **130**, and/or ATM **122** as will be described in greater detail below. An input device or component of the mobile I/O device **120** allows the customer **110** to provide information to the mobile device and may include, for example, a mechanical keyboard, a touchscreen, a microphone, a camera, a fingerprint scanner, any customer **110** input device engageable with the mobile device **114** via a USB, serial cable, Ethernet cable, and so on. An output device or component of the mobile I/O device **120** allows the customer **110** to receive information from the mobile device and may include, for example, a digital display, a speaker, illuminating icons, LEDs, and so on. Further, the mobile I/O device **120** may include assemblies that serve both input and output functions, allowing, for example, the ATM **122** and the financial institution computing system **130** to exchange information with the mobile device **114**. Such assemblies include, for example, radio frequency transceivers (e.g., RF or NFC-based transceivers) and other short-range wireless transceivers (e.g., Bluetooth®, laser-based data transmitters, etc.).

(28) The mobile wallet client application **118** is communicably coupled to the financial institution computing system **130** (e.g., the rewards management circuit **134** and the customer database **138**) and is structured to permit management of the customer financial accounts and transactions. In some embodiments, the mobile wallet client application **118** may be incorporated with an existing application in use by the financial institution (e.g., a mobile banking application). In some arrangements, the mobile wallet client application **118** is a separate software application implemented on the mobile device **114**. The mobile wallet client application **118** may be downloaded by the mobile device **114** prior to its usage, hard coded into the memory of the mobile device or be a web-based interface application such that the mobile device **114** may provide a web browser to the application, which may be executed remotely from the mobile device **114**. In the latter instance, the customer **110** may have to log onto or access the web-based interface before usage of the applications. Further, and in this regard, the mobile wallet client application **118** may be supported by a separate computing system including one or more servers, processors, network interface circuits, etc. that transmit applications for use to the computing device. In certain embodiments, the mobile wallet client application **118** includes an API and/or a software development kit (SDK) that facilitate the integration of other applications with the mobile wallet client application **118**.

(29) Irrespective of the form that the mobile wallet client application, the mobile wallet client application **118** is structured to enable the customer **110** to create, manage, and interact with a mobile wallet account. While setting up a mobile wallet account, the mobile wallet client application **118** may receive, organize, and store payment tokens from card network computing system **140**. The mobile wallet client application **118** may then transmit certain payment tokens to ATM **122** (e.g., via the mobile I/O device **120**) as part of any of a variety of transaction requests to complete various financial operations. For example, the mobile wallet client application **118** may receive and display screens from a computing system that operates a mobile wallet platform (e.g., the data exchange circuit **136** at the financial institution computing system **140**) on the mobile I/O device **120** including account information, transaction instructions, and so on. In one embodiment, a screen may be used to request a customer name, password, and PIN information from the customer as well as information sufficient to identify a target account (e.g., to identify a payment token associated with the credit card **112**) and a desired transaction type (e.g., a reward

withdrawal).

(30) An illustrative implementation of the reward redemption system **100** including the mobile wallet client application **118** can be described as follows. The customer **110** downloads and installs the mobile wallet client application **118** on the mobile device **114**. The mobile wallet client application **118** is communicatively coupled to the data exchange circuit **136** of the financial institution computing system **130** which is configured to provide the customer **110** with displays including all accounts held by the customer **110** at the financial institution. Accordingly, the customer **110** is presented with a display that includes a depiction of an account associated with the credit card **112**. The customer **110** then selects the account associated with the credit card **112** and the mobile wallet client application **118** is configured to communicate the selection to the financial institution computing system **130**. Responsive to the customer selection, the data exchange circuit **136** transmits account identifying information (e.g., a PAN) to the card network computing system **140** which generates (e.g., via the token provisioning circuit **144**) a payment token for the customer **110** and communicates the token back to the financial institution computing system **130** over the network **150**. It should be noted that, as described herein, in certain arrangements, the data exchange circuit **136** may perform all of the functions described herein with reference to the card network computing system **140**. The data exchange circuit **136** then transmits the generated token to the mobile device **114**.

(31) After the token information is stored on the mobile device **114**, the customer **110** then walks up to the ATM selects the payment token via the mobile wallet client application **118** and causes the mobile device **114** to transfer the payment token to the ATM **122**. Transferring the token may be accomplished through complementary contactless data transfer devices at the ATM **122** and the mobile device **114**. For example, an NFC device included in the mobile I/O device **120** can be used to wirelessly transfer the token to a receiving NFC device included in the ATM I/O device **126**. In some arrangements, the transmission of the payment token “wakes up” the ATM **122** from a sleep or standby mode (e.g., where a display at the ATM **122** passively displays a screensaver or a welcome or instruction screen). Responsive to receiving the token, the ATM circuit **126** may be configured to request additional information from the customer and transmit information to the financial institution computing system **130** for authenticating the customer, as will be described in greater detail below.

(32) Referring now to FIG. 2, a flow diagram of a method **200** for redeeming financial rewards is shown according to an example embodiment. The method **200** may be performed by processing and storage hardware at the financial institution computing system **130** (e.g., the financial institution computing system **130**), as executed by one or more circuits (e.g., the data exchange circuit **136** and the rewards management circuit **134**) configured to perform the functions described below.

(33) Customer account information is received at **202**. In various arrangements, the financial institution computing system **130** may receive the information from the ATM **122** over the network **150**. In some arrangements, the information includes account identifying information associated with the customer **110**. In some arrangements, the customer **110** inputs the information into the ATM **122** through the ATM I/O device **128**. For example, the customer **110** may insert the credit card **112** into the ATM I/O device **128** (e.g., through the card reader **129**). Upon receiving the credit card **112**, the ATM I/O device **128** may be configured to transmit an account number (e.g., a PAN) to the ATM circuit **126**, which is configured to determine that the account number is associated with the financial institution and transmit the account identifying information to the financial institution computing system **130** over the network **150**. In some arrangements, the ATM circuit **126** is further configured to present the customer **110** with displays on the ATM I/O device **128** instructing the customer to input authentication information (e.g., a PIN associated with the credit card **112**). Authentication information input by the customer **110** is then communicated to the ATM circuit **126** which packages the information with the account number to create a transaction request,

which is transmitted to the financial institution computing system **130** via the network interface circuit **124**.

(34) In some arrangements, the customer authentication credentials are received through a combination of the ATM I/O device **128** and the mobile device **114**. For example, the mobile device **114** may be preconfigured with a mobile wallet client application **118** and thus have a payment token corresponding to the customer account stored thereon. The customer **110** approaches the ATM **122** and allows the mobile device **114** (e.g. by NFC data transmission) to communicate with the ATM circuit **126** via the mobile I/O device **120** of the mobile device **114** and the ATM I/O device **128**. The payment token may then be transferred to either the financial institution computing system **130** or the card network computing system **140** for further processing, as discussed above. To provide an illustrative example, in one arrangement, the card network computing system **140** then detokenizes the payment token (e.g., using token mapping data stored in the token database **146**), identifies the PAN of the customer, and transmits the credentials to the financial institution computing system **130** for processing.

(35) It is determined if the received account information is associated with a credit card having financial rewards associated therewith at **204**. In some arrangements, the rewards management circuit **134** receives the account identifying information transmitted by the ATM **122** via the network interface circuit **132**, retrieves information associated with the account from the customer database **138**, and determines whether there are any financial rewards associated with the account identifying information.

(36) If there are no rewards associated with the input account identifying information, the customer **110** is presented with a credit card offer interface at **206**. For example, upon the rewards management circuit **134** determining that the received account information has no financial rewards, the rewards management circuit **134** may be further configured to transmit a credit card application interface to the ATM **122** for review by the customer **110** via the ATM I/O device **128**. In another arrangement, rather than presenting the customer **110** with an application interface, the customer **110** is presented with advertisements notifying the customer **110** of the financial institution available credit cards and associated financial reward programs.

(37) The customer **110** is authenticated at **208**. In some arrangements, where, for example, only account identifying information is received, the rewards management circuit **134** is configured to transmit an authentication information request to the ATM **122** over the network **150**. Upon receipt of the authentication information request, the ATM circuit **126** is configured to present the customer with a display on the ATM I/O device **128** prompting the customer to input authentication information (e.g., a PIN) associated with the received account information at **202**. Authentication information input by the customer via the ATM I/O device **128** is communicated to the ATM circuit **126** which may in turn transmit the authentication information to the financial institution computing system **130** over the network **150**. The rewards management circuit **134** receives the authentication information and determines whether the authentication information entered by customer matches credentials associated with the account stored in the customer database **138**.

(38) It is determined if the credit card is registered for an ATM cash rewards program at **210**. In some arrangements, the customer's registration status is stored in the customer database **138** and the rewards management circuit **134** is configured to retrieve the customer's registration status. If the customer **110** has not registered, then the rewards management circuit **134** may be configured to initiate a credit card registration process at **212** that will be described in greater detail below with reference to FIG. 3. If, however, the credit account is registered, then the method **200** continues.

(39) The customer's available reward balance is determined at **214**. In one arrangement, customer financial reward information is maintained at the customer database **138** by the rewards management circuit **134** at the financial institution computing system **130**. The rewards management circuit **134** is configured to retrieve customer financial reward information and determine whether the customer **110** has earned any financial rewards through utilization of

account associated with the credit card **112**. If the customer has earned rewards, the rewards management circuit **134** may be further configured to convert whatever form that the reward normally takes (e.g. points or airline miles) into a currency value in accordance with parameters set by the financial institution.

(40) A reward notification is presented to the customer **110** at **216**. In some arrangements, the reward notification is generated by the rewards management circuit **134** responsive to the customer reward determined at step **214**. The reward notification includes a graphical interface that is transmitted to the ATM **122** or the mobile device **114** over the network **150**. In one arrangement, the reward notification indicates whatever rewards that the customer has earned through utilization of the account associated with the credit card **112** and further indicates the cash value of any such earned rewards to the customer **110**. This reward notification may be relayed via the network **150** to the ATM **122** for display via the ATM I/O device **128**. In some arrangements, the reward notification also contains graphics giving the customer **110** the ability to indicate whether they wish to redeem the earned rewards (or any portion thereof) for cash.

(41) It is determined if the customer **110** redeemed any rewards at **218**. In some arrangements, the ATM I/O device **128** receives reward redemption information input by the customer **110** on the reward notification generated at **216**. In some arrangements, the ATM circuit **126** is configured to assess the reward redemption information to determine if the customer **110** indicated a wish to redeem a portion of the reward balance determined at **214**. In some arrangements, the ATM circuit **126** is configured to transmit the received reward redemption information to the financial institution computing system **130** over the network **150** and the reward management circuit **134** is configured to determine if the customer **110** indicated a wish to redeem financial rewards. In some arrangements, if the customer **110** did not indicate a wish to redeem rewards, the method **200** ends.

(42) A customer-preferred redemption amount is determined at **220**. In some arrangements, if the customer **110** indicates a reward redemption preference, the ATM circuit **126** is configured to formulate a transaction request reflecting the portion of the financial rewards that the customer **110** indicated that they wish to redeem. The ATM circuit **126** may then transmit this transaction request over the network **150** to the financial institution computing system **130**, where the data exchange circuit **136** or rewards management circuit **134** of the financial institution computing system **130** processes the transaction request to determine the redemption amount. The transaction request generated may include a specified transaction type (in this instance, a cash withdrawal), one or more corresponding transaction parameters (e.g., an amount to be withdrawn), and the specified financial account (e.g. the financial account through which the customer **110** was authenticated at **208**).

(43) Rewards are dispensed to the customer at **222**. In some arrangements, the data exchange circuit **136** completes the transaction based on the transaction request received at **220** and information in the customer database **138**. For example, where the transaction request includes a withdrawal of all earned financial rewards, the data exchange circuit **136** accesses reward balance information associated with the credit account linked to the account identified in the customer database **138**, adjusts bookkeeping entries, and transmits an approval and confirmation to the ATM **122** over the network **150**. Responsive to the approval and confirmation, the ATM circuit **126** communicates with circuitry associated with a cash dispenser included in the ATM I/O device **128**, and an amount of currency corresponding to that in the transaction request is dispensed to the customer **110**.

(44) Referring now to FIG. 3, a method **300** of registering a customer payment account for an ATM cash redemption program is shown according to an example embodiment. The method **300** is performed by processing and storage hardware at the financial institution computing system **130**. It should be understood that the initiation of the method **300** may take a variety of forms. For example, method **300** may be initiated responsive to the financial institution computing system **130** determining that the account associated with the credit card **112** of the customer **110** is not

registered (for example, at **210** in the method **200** above).

(45) In some arrangements, the method **300** may be initiated responsive to the customer **110** indicating a wish to register a credit card **112** for an ATM cash redemption program during an ATM session at the ATM **122**. The ATM session may be associated with another payment account held by the customer **110** at the financial institution, or it may be a session engaged in the customer **110** solely for the purpose of registering the credit card for the ATM cash rewards program. To provide an illustrative example, the customer **110** may have initiated an ATM session using a debit card issued by the financial institution. In the session, the customer may have indicated (e.g., via the ATM I/O device **128**) a preference to register a different payment account (e.g., the one associated with the credit card **112**) for an ATM cash redemption program. Responsive to this preference, the ATM circuit **126** is configured to generate instructions, displayed through the ATM I/O device **128** corresponding to the method **300**.

(46) A registration input is received at **302**. In some arrangements, this registration input may include the receipt of account identifying information or customer credentials corresponding to a credit card not registered for the ATM cash redemption program (e.g. in the steps **202-208** discussed above relating to the method **200**). In some arrangements the registration input includes credit account identifying information (e.g. a credit card number) received from either the ATM I/O device **128** or the mobile device **114**. In some arrangements, the registration input is received during an ATM session at the ATM **122** initiated through payment account other than the account associated with the credit card **112** (e.g., a debit card). Upon receipt of the account identifying information, the ATM circuit **126** is configured to relay this information to the financial institution computing system **130** for assessment by the rewards management circuit **134**, which determines whether the account information received is associated with a credit card rewards account and, if so, generates graphical interfaces to be relayed to the ATM **122** for display to the customer **110** for carrying out the additional steps of the method **300** described below. In some arrangements, the registration input may take the form of a customer input received from the ATM I/O device **128** indicating a desire to register a credit card.

(47) It is determined if the customer **110** has already registered the account associated with the credit card **112** at **304**. In some arrangements, based on the account identifying information received at **302**, the financial institution computing system **130** uses similar circuitry and methodologies as discussed in relation to the step **204** of method **200** discussed in relation to FIG. 2. In some arrangements, if the customer **110** has already registered the credit account, then the rewards management circuit **134** is configured to transmit a notification to the ATM **122** for presentation to the customer. In some arrangements, responsive to it being determined that the customer **110** has already registered the credit account, the rewards management circuit **134** is configured to initiate the method **200** discussed above at **306** to enable the customer **110** to redeem any earned financial rewards for cash. If the customer has not registered the credit account for the ATM cash redemption program, then the method **300** continues.

(48) The customer is authenticated at **308**. In some arrangements, when the registration input received at **302** includes both customer credentials (e.g., credentials associated with another payment account, such as a debit card, of the customer **110**) and credit account identifying information, the rewards management circuit **134** determines whether the customer credentials (e.g., a PIN entered by customer **110** associated with the debit account via the ATM I/O device **128**) matches customer credentials stored in the customer database **138**. In some arrangements, where the registration input includes solely credit account identifying information, for example, authentication requires additional information from the customer **110**. For example, responsive to just receiving account identifying information (i.e., when the customer **110** simply enters the credit card **112** into the ATM I/O device **128**), the rewards management circuit **134** is configured to transmit instructions to the ATM **122**, which may configure the ATM circuit **126** to present displays to the customer **110** via the ATM I/O device **128** that prompt the customer **110** to input customer-

specific information (e.g., date of birth, social security number, etc.) that can be cross referenced with information stored in the customer database **138** for authentication. In some arrangements, the ATM circuit **126** receives the customer-specific information and transmits the information to the financial institution computing system **130**. The information may then be assessed by the rewards management circuit **134** to determine whether the received customer-specific information matches customer information stored in the customer database **138**.

(49) The customer **110** is presented with a registration interface at **310**. In some arrangements, responsive to the customer **110** being authenticated, the rewards management circuit **134** is configured to generate a registration interface prompting the customer **110** to input registration information. The displays may be transmitted to the ATM **122** over the network **150** and presented to the customer **110** via the ATM I/O device **128**. In some arrangements, the registration information requested from the customer **110** may include credit account identifying information. For example, if credit account identifying information has not yet been obtained (e.g., from the actual credit card **112** itself or the mobile device **114** through a mobile wallet client application **118**), the registration interface may include instructions prompting the customer **110** to insert the credit card **112** into the ATM I/O device **118** or to tap the mobile device **114** to transmit a token. In some arrangements, the registration interface instructs the customer **110** to establish authentication credentials to be associated with the credit account. For example, the registration information may request the customer **110** to establish a PIN that can be used to authenticate the customer **110**. In some arrangements, the registration interface may also include a registration confirmation query, which requests a customer **110** to confirm a preference of the customer **110** to register the credit account for the ATM cash redemption program.

(50) In some arrangements, the registration interface instructs the customer **110** to indicate if they have a preference to identify any additional credit accounts for the ATM cash redemption program. In some arrangements, the rewards management circuit **134** is configured to identify additional credit accounts associated with the customer **110** based on information stored in the customer database **138**, and the registration interface identifies the additional credit accounts and provides the customer **110** with the ability to indicate a preference to register at least one of the additional credit accounts for the ATM cash redemption program. In some arrangements, the registration interface identifies the financial reward programs that are associated with the credit card **112**.

(51) Customer registration information is received at **312**. In some arrangements, the customer **110** inputs the information requested by the registration interface generated at **310** via the ATM I/O device **128**. The ATM circuit **126** receives the input information, and transmits the input information to the financial institution computing system **130** over the network **150**.

(52) The registration information is associated with a customer account at **314**. In some arrangements, the rewards management circuit **134** is configured to store the registration information received at **312** in the customer database **138** in association with the credit account of the customer.

(53) It is determined if the customer **110** indicated a preference to register additional credit cards for the ATM cash program at **316**. In some arrangements, the rewards management circuit **134** is configured to assess the customer registration information received at **312** to determine if the customer **110** indicated such a preference. In some arrangements, after the customer **110** has registered a first credit account for the ATM cash withdrawal program, the rewards management circuit **134** is configured to identify additional credit accounts that the customer **110** has and to transmit an additional registration interface identifying the additional credit account to the ATM **122** for presentation to the customer **110**.

(54) If the customer **110** has indicated a preference to register additional credit accounts for the ATM cash redemption program, then the ATM circuit **126** may be configured to revert back to **316** and generate a registration interface prompting the customer **110** to input any additional information necessary (using the systems and methods discussed above) to register a second credit account for

the ATM cash redemption program. If, however, the customer **110** has not indicated any preference to register additional credit accounts, then the ATM circuit **126** may be configured to initiate the method **200** at **306** to enable the customer to redeem any earned financial rewards associated with the newly registered credit account.

(55) Referring now to FIG. **4A**, an interface **400** on a display included in the ATM I/O device **128** is shown. In some arrangements, the interface **400** is presented after the customer **110** inputs authentication information (e.g., at the step **202** at the method **200** displayed in FIG. **2**) via the ATM I/O device **128**. As shown, the interface **400** includes a reward redemption button **402**, a reward balance inquiry button **404**, a register additional cards button **406**, and a more options button **408**. Selecting the redeem available awards button **402** allows the customer **110** to select a financial reward associated with the credit card **112** and to exchange currency for any portion of the selected financial reward. Selecting the reward balance inquiry button **404** allows the individual to receive information relating to account various financial awards that the customer **110** has in relation to the credit account. Selecting the register additional cards button **406** allows the customer **110** to register additional credit accounts (e.g., by the process **300** displayed in FIG. **3**) for the ATM cash redemption program. Selecting the more options button **408** allows the customer **110** to access additional functions of the ATM **122** that are not shown on the interface **400**. One option could be for the customer **110** to pair the mobile device **114** with the ATM **122**.

(56) In FIG. **4B**, an interface **410** on a display included in the ATM I/O device **128** is shown according to an example embodiment. In some arrangements, the interface **410** is presented after the customer **110** selects an option to pair the mobile device **114** to the ATM **122** after hitting the other options button **408** displayed in FIG. **4A**. The interface **410** includes an NFC tap to pair button **412**, a Bluetooth® tap to pair button **414**, a scan code to pair button **416**, and a cancel button **418**. The cancel button **418**, in one arrangement, causes the ATM **122** to return to the start screen **400** displayed in FIG. **4A**. The NFC tap to pair button **412** causes the ATM **122** to establish a connection through a NFC transceiver to the mobile device **114** for data communication between the ATM **122** and the mobile device **114** (e.g., the transmittal of a payment token to the ATM **122**). The Bluetooth® tap to pair button **414** causes the ATM **122** to establish a communication through a Bluetooth® connection to the mobile device **114**. The scan code to pair button **416** causes the ATM **122** to display a barcode (e.g., a QR code) that can be scanned by a camera associated with the mobile device **114**. In some arrangements, the barcode launches a website to be used as a secure data portal for communication with the ATM **122** or the financial institution computing system **130**.

(57) Referring now to FIG. **4C**, an interface **420** on a display included in the ATM I/O device **128** of an ATM **122** is shown according to an example embodiment. In some arrangements, the interface **420** is displayed based on the financial institution computing system **130** determining that the customer **110** has registered the credit card **112** for the ATM cash redemption program and that the customer **110** has available rewards to redeem (e.g., at **216** in the method **200** displayed in FIG. **2**). Alternatively, the interface **420** is displayed responsive to the customer **110** selecting the reward balance inquiry button **404** displayed in FIG. **4A**. The interface **420** includes a window **422** having a reward notification **424**, a complete reward redemption button **426**, a portion reward redemption button **428**, and a reward denial button **430**. The reward indication **424** indicates a cash value of earned financial rewards of the customer. The reward indication **424** can also include other values (points, airfare mileage, etc.) indicating the total amount of earned financial rewards. The complete reward redemption button **426** enables the customer **110** to exchange the indicated amount of currency for earned financial rewards. The portion rewards redemption button **428** enables the customer **110** to input a portion of the indicated value of the earned by the customer **110**. In some arrangements, the portion rewards redemption button **428** enables the customer **110** to indicate any cash value that is less than that indicated by the reward indication **424**. Upon customer selection of this button, a notification may be relayed to the financial institution computing system **130** for processing by the rewards management circuit **134** to update the reward account balance of the

customer in a manner that correlates with the portion of the reward that the customer **110** chose to redeem for cash at the ATM **122**. The reward denial button **430** enables the customer **110** to not redeem any of the earned financial rewards at the ATM **122**.

(58) Referring now to FIG. **4D**, an interface **432** on a display included in the ATM I/O device **128** of an ATM **122** is shown according to an example embodiment. In some arrangements, the interface **432** may be shown to the customer **110** upon selection of the register additional cards button **406** displayed in FIG. **4A**. In some arrangements, the interface **432** may be presented to the customer as the registration interface (e.g., in the step **310** of the method discussed above in relation to FIG. **3**). The interface **432** includes a generate customer PIN button **434**, a reward account selection button **436**, a pair mobile device button **438**, and a cancel button **440**. The cancel button **440** cancels the registration methods described herein. The reward account selection button **436** allows the customer **110** to select a particular financial reward program to register for ATM cash withdrawals. Upon selection of the reward account selection button **436**, the customer **110** may be further brought to another interface displaying all of the customer financial reward information (retrieved from, e.g., the customer database **138** via the rewards management circuit **134**) so as to enable the customer **110** to choose a particular reward account. The generate customer PIN button **438** enables the customer **110** to layer an additional layer of authentication onto any unregistered credit card. Selection of the generate customer PIN button **526** brings the customer **110** to a screen prompting the customer **110** to input and confirm a PIN (e.g., via a keyboard of the ATM I/O device **128**) to be paired with the unregistered credit card. The confirmed pin will be later used to authenticate the customer **110** in making cash reward withdrawals (e.g., in the methods **200** and **300**). The pair mobile device **438** enables the customer to communicatively couple the mobile device **114** with the ATM **122** (using any relevant methodologies discussed above) so that the customer **110** can select a credit card that they wish to register for the ATM cash reward program via the mobile wallet client application **118** so that tokenized identifying information associated with the card can be transmitted to the financial institution computing system **130** for registration. Selection of the pair mobile device button **528** may bring to the interface discussed in FIG. **4B** above, enabling the customer **110** to use multiple methods to pair the mobile device **114** to the ATM **122**.

(59) The embodiments described herein have been described with reference to drawings. The drawings illustrate certain details of specific embodiments that implement the systems, methods and programs described herein. However, describing the embodiments with drawings should not be construed as imposing on the disclosure any limitations that may be present in the drawings.

(60) It should be understood that no claim element herein is to be construed under the provisions of 35 U.S.C. § 112(f), unless the element is expressly recited using the phrase “means for.”

(61) As used herein, the term “circuit” may include hardware structured to execute the functions described herein. In some embodiments, each respective “circuit” may include machine-readable media for configuring the hardware to execute the functions described herein. The circuit may be embodied as one or more circuitry components including, but not limited to, processing circuitry, network interfaces, peripheral devices, input devices, output devices, sensors, etc. In some embodiments, a circuit may take the form of one or more analog circuits, electronic circuits (e.g., integrated circuits (IC), discrete circuits, system on a chip (SOCs) circuits, etc.), telecommunication circuits, hybrid circuits, and any other type of “circuit.” In this regard, the “circuit” may include any type of component for accomplishing or facilitating achievement of the operations described herein. For example, a circuit as described herein may include one or more transistors, logic gates (e.g., NAND, AND, NOR, OR, XOR, NOT, XNOR, etc.), resistors, multiplexers, registers, capacitors, inductors, diodes, wiring, and so on). The term “circuit” may also include one or more dedicated processors communicatively coupled to one or more dedicated memory or memory devices. In this regard, the one or more processors may execute instructions stored in the memory or may execute instructions otherwise accessible to the one or more

processors. In some embodiments, the one or more processors may be embodied in various ways. The one or more processors may be constructed in a manner sufficient to perform at least the operations described herein. In some embodiments, the one or more processors may be shared by multiple circuits (e.g., circuit A and circuit B may comprise or otherwise share the same processor which, in some example embodiments, may execute instructions stored, or otherwise accessed, via different areas of memory). Alternatively or additionally, the one or more processors may be structured to perform or otherwise execute certain operations independent of one or more co-processors. In other example embodiments, two or more processors may be coupled via a bus to enable independent, parallel, pipelined, or multi-threaded instruction execution. Each processor may be implemented as one or more general-purpose processors, application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), digital signal processors (DSPs), or other suitable electronic data processing components structured to execute instructions provided by memory. The one or more processors may take the form of a single core processor, multi-core processor (e.g., a dual core processor, triple core processor, quad core processor, etc.), microprocessor, etc.

(62) An example system for implementing the overall system or portions of the embodiments might include a general purpose computing computers in the form of computers, including a processing unit, a system memory, and a system bus that couples various system components including the system memory to the processing unit. Each memory device may include non-transient volatile storage media, non-volatile storage media, non-transitory storage media (e.g., one or more volatile and/or non-volatile memories), etc. In some embodiments, the non-volatile media may take the form of ROM, flash memory (e.g., flash memory such as NAND, 4D NAND, NOR, 4D NOR, etc.), EEPROM, MRAM, magnetic storage, hard discs, optical discs, etc. In other embodiments, the volatile storage media may take the form of RAM, TRAM, ZRAM, etc. Combinations of the above are also included within the scope of machine-readable media. In this regard, machine-executable instructions comprise, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions. Each respective memory device may be operable to maintain or otherwise store information relating to the operations performed by one or more associated circuits, including processor instructions and related data (e.g., database components, object code components, script components, etc.), in accordance with the example embodiments described herein.

(63) It should also be noted that the term “input devices,” as described herein, may include any type of input device including, but not limited to, a keyboard, a keypad, a mouse, joystick or other input devices performing a similar function. Comparatively, the term “output device,” as described herein, may include any type of output device including, but not limited to, a computer monitor, printer, facsimile machine, or other output devices performing a similar function.

(64) Any foregoing references to currency or funds are intended to include fiat currencies, non-fiat currencies (e.g., precious metals), and math-based currencies (often referred to as cryptocurrencies). Examples of math-based currencies include Bitcoin, Litecoin, Dogecoin, and the like.

(65) It should be noted that although the diagrams herein may show a specific order and composition of method steps, it is understood that the order of these steps may differ from what is depicted. For example, two or more steps may be performed concurrently or with partial concurrence. Also, some method steps that are performed as discrete steps may be combined, steps being performed as a combined step may be separated into discrete steps, the sequence of certain processes may be reversed or otherwise varied, and the nature or number of discrete processes may be altered or varied. The order or sequence of any element or apparatus may be varied or substituted according to alternative embodiments. Accordingly, all such modifications are intended to be included within the scope of the present disclosure as defined in the appended claims. Such

variations will depend on the machine-readable media and hardware systems chosen and on designer choice. It is understood that all such variations are within the scope of the disclosure. Likewise, software and web implementations of the present disclosure could be accomplished with standard programming techniques with rule based logic and other logic to accomplish the various database searching steps, correlation steps, comparison steps and decision steps.

(66) The foregoing description of embodiments has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure to the precise form disclosed, and modifications and variations are possible in light of the above teachings or may be acquired from this disclosure. The embodiments were chosen and described in order to explain the principals of the disclosure and its practical application to enable one skilled in the art to utilize the various embodiments and with various modifications as are suited to the particular use contemplated. Other substitutions, modifications, changes and omissions may be made in the design, operating conditions and arrangement of the embodiments without departing from the scope of the present disclosure as expressed in the appended claims.

Claims

1. An automated teller machine (ATM) comprising: a network interface configured to communicate with one or more remote computing systems through a telecommunications network; an input/output device configured to exchange data with a user, the input/output device comprising a card reader, a display screen, and a cash dispenser; and a processing circuit comprising a processor and memory, the memory structured to store instructions that are executable by the processor and cause the processing circuit to: receive, via the input/output device, information identifying an account associated with the user; determine, based on the information, that the account associated with the user is not registered for a cash redemption program corresponding to one or more rewards accounts associated with the user; upon determining that the account is not registered for the cash redemption program, present, via the input/output device, a rewards selection graphical user interface to select rewards accounts for registration in the cash redemption program, the rewards selection graphical user interface comprising the one or more rewards accounts associated with the user, the one or more rewards accounts different from the account associated with the user; responsive to a selection of a first rewards account of the one or more rewards accounts displayed at the rewards selection graphical user interface, register the first rewards account for the cash redemption program; display, via the input/output device, a second graphical user interface comprising a plurality of reward indications that includes (i) a first option to redeem a currency amount corresponding to a total value of rewards indicated in rewards information received via the network interface, and (ii) a second option to redeem a user-specified amount of currency corresponding to a respective portion of the total value of rewards indicated in the rewards information, each of the first option and the second option providing the user with an ability to perform a currency withdrawal based on the rewards information; detect, via the input/output device, an input by the user into the second graphical user interface, the input indicating a selection of the second option to perform the currency withdrawal; detect, via the input/output device, an additional input by the user that specifies a portion of the total value of rewards indicated in the rewards information to withdraw as cash; transmit, via the network interface, a reward redemption request based on the selection and the additional input; receive, via the network interface, a reward redemption authorization; and dispense, via the input/output device, a cash amount corresponding to the portion of the total value in cash in response to receiving the reward redemption authorization.
2. The ATM of claim 1, wherein the processing circuit is further configured to: display, via the input/output device in response to determining that the account is not registered for the cash redemption program, a first graphical user interface comprising a list of interactive user interface

elements, (i) a first interactive user interface element of the list, when interacted with, providing the user with an ability to establish a secure identifier for the account, and (ii) a second interactive user interface element of the list, when interacted with, causing the ATM to present the rewards selection graphical user interface.

3. The ATM of claim 1, wherein the processing circuit is further configured to update the total value based on the portion.

4. The ATM of claim 1, wherein the processing circuit is further configured to display a selectable option to pair a mobile device in registering for the cash redemption program.

5. The ATM of claim 1, wherein the processing circuit is further configured to: receive data indicative of an interaction of the user with the ATM through the input/output device; and initiate a user ATM session responsive to the data indicative of the interaction.

6. The ATM of claim 5, wherein the data indicative of the interaction is tokenized and received from a computing device associated with the user.

7. The ATM of claim 5, wherein the data indicative of the interaction includes an account number associated with the account.

8. The ATM of claim 1, wherein the processing circuit is further configured to: display an instruction for an input comprising authentication information; receive the authentication information; and generate a transaction request comprising the authentication information and the account information.

9. The ATM of claim 1, wherein the processing circuit is further configured to present a plurality of buttons, the plurality of buttons comprising: a reward redemption button structured to commence a reward redemption process; a reward balance inquiry structured to present financial award information; and a register additional cards button structured to commence an account registration process.

10. The ATM of claim 1, wherein the processing circuit is further configured to register the account associated with the user.

11. The ATM of claim 1, wherein the account is a credit account.

12. A computer-implemented method comprising: receiving, by an automated teller machine (ATM) comprising a network interface and an input/output device, account information identifying an account associated with a user; responsive to receiving the account information, determining, by the ATM, based on rewards information received via the network interface, that the account is not registered for a cash redemption program corresponding to one or more rewards accounts of the user; responsive to determining that the account is presenting, by the ATM via the input/output device a rewards selection graphical user interface to select at least one of the one or more rewards accounts for registration in a cash redemption program, the rewards selection graphical user interface comprising the one or more rewards accounts associated with the user, the one or more rewards accounts different from the account associated with the user; responsive to an interaction with the rewards selection graphical user interface, displaying, by the ATM via the input/output device, the rewards selection graphical user interface comprising the one or more rewards accounts associated with the user; responsive to a selection of a first rewards account of the one or more rewards accounts displayed at the rewards selection graphical user interface, registering, by the ATM, the first rewards account for the cash redemption program; responsive to an interaction with the rewards selection graphical user interface, displaying, by the ATM, a second graphical user interface providing a plurality of reward indications that includes (i) a first option to redeem a currency amount corresponding to a total value of rewards indicated in rewards information received via the network interface, and (ii) a second option to redeem a user-specified amount of currency corresponding to a respective portion of the total value of rewards indicated in the rewards information, each of the first option and the second option providing the user with an ability to perform a currency withdrawal based on the rewards information; detecting, via the input/output device, an input by the user into the second graphical user interface, the input

indicating a selection of the second option to perform the currency withdrawal; detecting, via the input/output device, an additional input by the user that specifies a portion of the total value of rewards indicated in the rewards information to withdraw as cash; transmitting, by the ATM, based on the selection, a reward redemption request to a remote computing system and the additional input; receiving, by the ATM, a reward redemption authorization from the remote computing system; and dispensing, by the ATM, a cash amount corresponding to the portion of the total value in cash in response to receiving the reward redemption authorization.

13. The method of claim 12, further comprising: displaying, via the input/output device in response to determining that the account is not registered for the cash redemption program, a first graphical user interface comprising a list of interactive user interface elements, (i) a first interactive user interface element of the list, when interacted with, providing the user with an ability to establish a secure identifier for the account, and (ii) a second interactive user interface element of the list, when interacted with, causing the ATM to present the rewards selection graphical user interface.

14. The method of claim 12, further comprising updating, by the ATM, the total value based on the portion.

15. The method of claim 12, further comprising displaying, by the ATM, a selectable option to pair a mobile device in registering for the cash redemption program.

16. The method of claim 12, further comprising: receiving, by the ATM, data indicative of an interaction of the user with the ATM; and initiating, by the ATM, a user ATM session responsive receiving the data indicative of the interaction.

17. The method of claim 16, wherein the data indicative of the interaction is tokenized and received from a mobile computing device associated with the user.

18. The method of claim 16, wherein the data indicative of the interaction includes an account number associated with the account.

19. The method of claim 12, further comprising: displaying, by the ATM, an instruction for an input comprising authentication information; receiving, by the ATM, the authentication information; and generating, by the ATM, a transaction request comprising the authentication information and the account information.

20. The method of claim 12, further comprising presenting, by the ATM, a plurality of buttons, the plurality of buttons comprising: a reward redemption button structured to commence a reward redemption process; a reward balance inquiry structured to present financial award information; and a register additional cards button structured to commence an account registration process.
